



NUCLEAR ATOMS

Nuclear atom:

Types of Nuclear/atomic particles:

- 1) Electron : Charge, -1.6×10^{-19} C, Mass, 9.11×10^{-31} kg
- 2) Proton : Charge, $+1.6 \times 10^{-19}$ C, Mass, 1.66×10^{-27} kg
- 3) Neutron : Charge, Zero, Mass, 1.66×10^{-27} kg

Atom is electrically neutral

Nuclear composition:



A = Nucleon no.

Z = Atomic no.

A - Z = Neutron no.

For atomic composition, add the no. of electrons.

Isotopes: Atoms of same element with equal no. of protons and different no. of neutron/nucleons

e.g:- ${}^1_1\text{H}$, ${}^2_1\text{H}$, ${}^3_1\text{H}$

Atomic model:

Thompson's model: Atom is made of cloud of positive charges with scattered electrons.

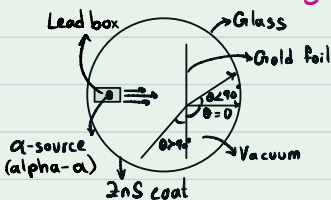


- Limitations:
- 1) No neutrons
 - 2) No electrons

Rutherford's model: Assumptions

- 1) Most of the atom is empty space.
- 2) All positive charges are concentrated in a small region called nucleus.
- 3) Nucleus is small and dense surrounded by equal no. of electrons as protons.

Rutherford's α -scattering experiment:



Observations:

1. Most α -particles go straight through the foil.
2. Few deviates with angle $< 90^\circ$.
3. Very few deviates with angle $> 90^\circ$.

* ZnS_2 used in TV screen

Conclusions

1. Most of Gold foil/atom is empty space.
2. There is a nucleus with concentration of charge.
3. Nucleus is +vely charged and it is small and dense.

1* Need of vacuum.

Alpha particles can collide with air particles and deflect before reaching the gold foil. So experiment is conducted in vacuum. They may not reach the foil.

2* Need of lead box.

Lead absorbs α -particles. This ensures that α -particles only comes out through the slit in the box. This confirms the initial direction of alpha particles.

3* Need of ZnS coat.

Alpha particles hitting ZnS coat creates bright spot which can be observed easily to find final position of the particle.

4* Need of same kinetic energy of α -particles.

Alpha particles with different k.e will have different amount of deviation. So k.e has to be controlled.

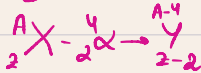
$$\text{K.E.} = \frac{1}{2}mv^2$$

RADIOACTIVITY

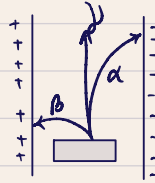
Types of radioactive particles:-

Alpha: (α) * Positively charged Helium nucleus. (α -nature/definition)

- * Highly ionizing.
- * Can be deflected by electric and magnetic field.
- * Can be stopped by thick paper.
- * Small range in air ($\approx 5-8\text{cm}$).
- * Small speed in air.
- * Decay equation:

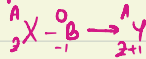


$$\alpha^{+2}, \beta^{+2}, \beta^{-1}, \gamma^0$$



Beta: (β) 1. Negatively charged fast electron (${}^0_{-1}e$)

2. Less ionizing than α .
3. Faster than α .
4. Can be stopped by Aluminium.
5. Can be deflected by electric and magnetic field.
6. Range in air about 1m.
7. Decay equation



Gamma rays: (γ)

- 1) High frequency EM wave.
- 2) Causes little/no ionization.
- 3) Infinite range in air. (compared with α and β)
- 4) Travels at speed of light in air.
- 5) Can be partly stopped by lead.
- 6) Not deflected by electric/magnetic field.

Radioactive emission: Random and continuous emission of radioactive particles from unstable nucleus.

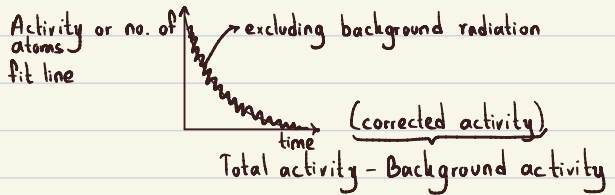
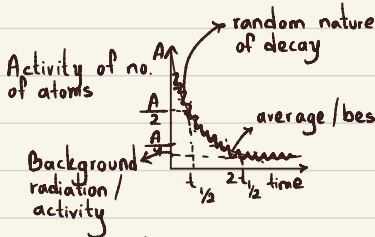
Activity: No. of disintegration per unit time unit Becquerel (Bq)

$$1 \text{ Bq} = 1 \text{ disintegration/sec.}$$

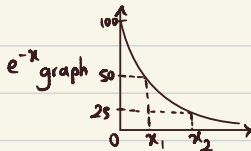
* Random means we cannot predict which nucleus will decay next.

* Spontaneous/continuous means radioactivity is unaffected by external factors.

Decay curve: (exponential graph)



* Each half value of ' y ' gives same value of ' x '.



$$x_1 = (x_2 - x_1)(x_3 - x_2)$$

Background activity: Amount of radiation present naturally in atmosphere.

Sources: Natural: Radon gas, underground rocks, cosmic rays etc.

Artificial: Nuclear power plant, medical equipments etc.

Half-life: Time taken for activity to drop to half of initial.

* A sample has activity 1000 bq and half life of 20 min. Find activity after 40 min.

$$40 \text{ min} = \frac{40}{20} = 2 \text{ half life}$$

$$\therefore \text{activity} = \frac{1}{2} \times \frac{1}{2} \times 1000 \\ = 250 \text{ bq}$$

* a) A substance has activity of 1000 bq and half life of 20 min. If background radiation is 100 bq, find the activity of source after 40 min.

b) What is the reading in detector after 40 min.

a) corrected activity = $1000 - 100 = 900 \text{ bq}$

$$\begin{aligned} 40 \text{ min} &= \frac{40}{20} = 2 \text{ half life} \\ \therefore \text{activity} &= \frac{1}{2} \times \frac{1}{2} \times 900 \\ &= 225 \text{ bq} \end{aligned}$$

b) Detected activity = $100 + 225 = 325 \text{ bq}$
reading of detector = 325 bq

Hazards of radiation:

- 1) Causes cell damage.
- 2) Can cause abnormal cell mutation which may lead to cancer.

Precautions:

- 1) Do not touch source with hands. Use tongs.
- 2) Wear protective apron.
- 3) Reduce exposure time.
- 4) Keep source inside lead box.
- 5) Do not drink/eat near radioactive sources.

Detection of Radiation/activity and Measurement:

Radioactivity is detected by cloud chamber, G-M (Geiger-Müller) tube etc.

Radiation is measured by G-M counter/rate meter.

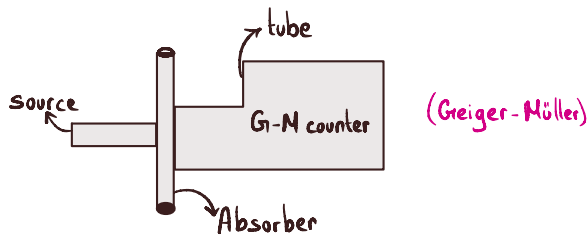
Use of G-M counter to detect type of radioactive source:

Switch on the counter without any source. Record the reading as background count rate.

Use the source inside G-M tube and read the reading. Now place thick paper between source and tube. If count rate drops to background rate, source is α -emitter.

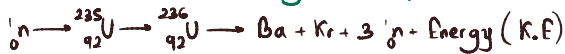
If not, Aluminium is used in place of paper. Drop in reading to background confirms Beta decay (β).

For no change with paper/Aluminium, lead is used as absorber. Radiation detected decreases which confirms Gamma decay (γ).



Nuclear fission:

Reaction where a massive/large nuclei splits into smaller ones by releasing energy.

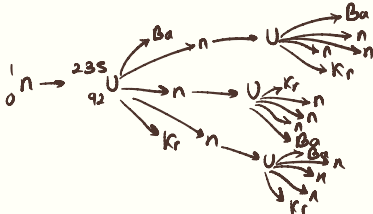


Mass of $(\text{n} + {}_{92}^{235}\text{U}) > (\text{Ba} + \text{Kr} + 3\text{n})$

* Mass defect converts into energy as $E = \Delta mc^2$ } $\Delta m = m_1 - m_2$

This energy is used in Nuclear power station to generate electricity

Chain reaction:



In this process released neutrons from initial fission are absorbed by other nuclei and more neutrons are released.

Components of Nuclear reactor:

Energy conversion: K.E from fission $\rightarrow$ Heat in water.

$\rightarrow$ K.E of steam $\rightarrow$ K.E in turbine.

$\rightarrow$ Electrical in generator.

1) Fission rod/fuel rod:

Fission occurs here. Mostly made of U-235.

2) Moderator:

Core of Reactor is surrounded by Graphite which acts as moderator. It absorbs K.E of neutrons and slow them down for further chain reaction.

3) Control rod:

Mostly made of Cadmium or Boron. These absorb neutrons and slow down or stop fission.

4) Coolant:

Water in pipes are run in and out of reactor core. Water carries heat from core and transfers energy outside.

5) Shield:

Reactor is surrounded by thick steel and concrete. These prevent/stops radiation from escaping out.

* Water from the core converts into steam. Steam pipe runs through container full of water. Water gains energy and steam condenses. Hot water turns into steam and runs a turbine to generate electricity.

Explain the hazards from reactor meltdown.

Reactor contains radioactive materials which emit ionising radiations. These radiations can damage cells, cause abnormal cell mutation leading to cancer. During meltdown radioactive particles escape in atmosphere. They spread over large area in a short time carried by wind. Everything within the area becomes contaminated, having a very long half-life. The area remains radioactive.

Fission: occurs when two or more nuclei collide or join each other to form a larger nuclei and release energy.

Example: $\underbrace{{}^2\text{H} + {}^2\text{H}}_{m_1} = \underbrace{{}^4\text{He}}_{m_2}$ Mass $m_1 > m_2$ ($E = \Delta mc^2$) * mass change converts to Energy
speed of light

In order to fuse, nuclei have to overcome their repulsive force. So they must have very high K.E.

This is done in stars by very high temperature and pressure.

In a fusion reactor, high K.E is reached by strong electric field and collision is caused by deflecting charges using magnetic field.

* Comparison between fission and fusion

Similarity: Both reactions provide energy from mass defect.

Disimilarity: 1. fission involves radioactive materials. fusion may not.

2. Energy of a single fusion is more than that of fission.

3. Fission requires change in speed of neutron for change in reaction. fusion depends on individual collision.

4. For fusion reactor electric and magnetic fields are needed whereas fission doesn't require these.

Applications of Radioactivity:-

1. Thickness monitoring.

2. Medical and non-medical tracer.

3. Radiocarbon dating.

4. Nuclear fission.

5. Nuclear fusion.

6. Finding leaks in pipe.

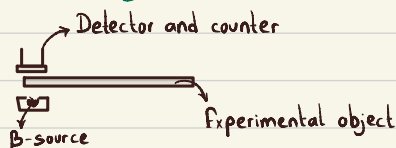
Radiocarbon dating:-

It is used to find age of dead bodies.

Atmosphere contains stable C-12 and radioactive C-14 ratio $\frac{C-14}{C-12}$. In atmosphere remains constant on average. Living bodies absorb C-12 and C-14 from atmosphere. So $\frac{C-14}{C-12}$ inside living body is same as surrounding. In a dead body C-14 continues to decay/decrease but C-12 remains same. So in a dead body sample $\frac{C-12}{C-14}$ changes in fraction compared to that outside. This fractional change is compared with half-life of C-14 provides approximately age of dead body.



Thickness monitoring:-



* α -particle is not used as it gets absorbed.

* γ ray will penetrate through

Beta radiation detected changes with change in thickness. This allows faults in production line to be detected.

Tracer (Medical):

A small amount of radioactive material is passed into the body. The material is absorbed or collected inside organs from where radiation is emitted.

By measuring amount of radiation count after certain time. Activity of organ can be compared with a healthy one.

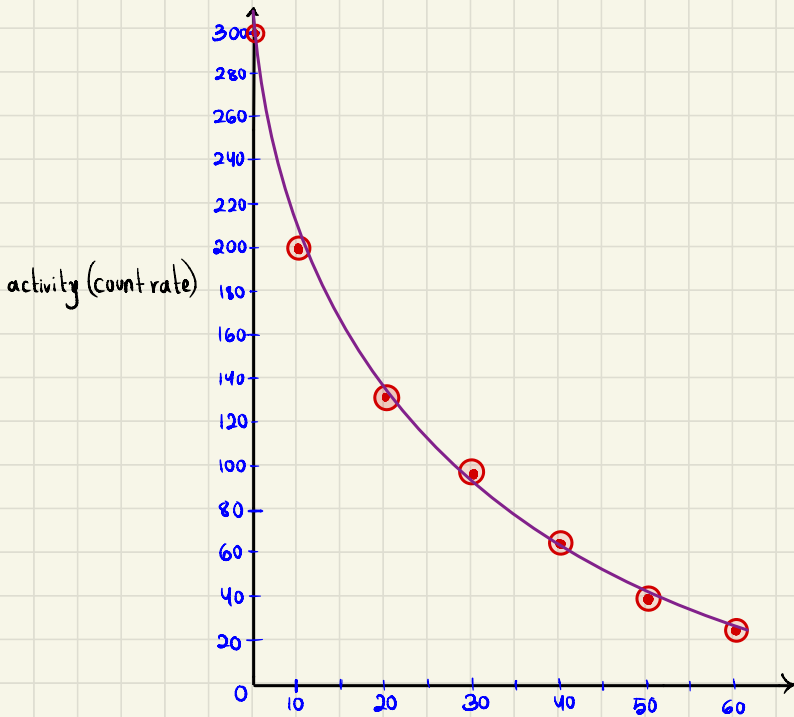
- 1) Type of emitter.
- 2) Amount (count rate)
- 3) Half life

Exponential graph - Radioactivity

Time/min	0	10	20	30	40	50	60
Count rate/ (counts/min)	330	230	165	120	92	70	56
Corrected radiation	300	200	135	90	62	40	26

Background radiation = 30 counts/min.

- Plot corrected activity-time graph.
- Find half life from graph.



	α	β	γ
Charge	Positive	Negative	No charge
Ionization	Strongest ionization	Less than α	Less than β
Penetration	Least	More than α	Most penetrating
Protection	A thick sheet of paper	A few millimetres of Perspex or aluminium	Several centimetres of lead
Deflection in electric field	Can be deflected	Can be deflected	Not deflected
Deflection in magnetic field	Can be deflected	Can be deflected	Not deflected